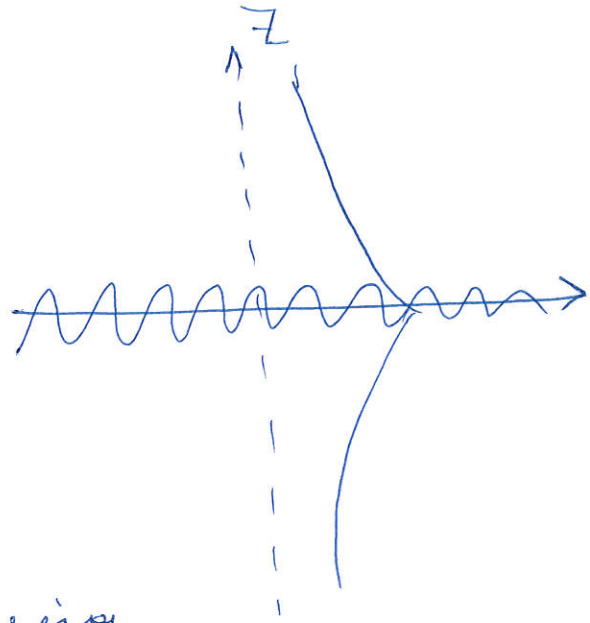
DISCUSSION – 10th March 2016

SURFACE PLASMONS (CONTD)



Assuming a particular form of $\vec{E}$ and $\vec{B}$ fields.

$$\vec{E} = \left(E_x(z),\ 0,\ E_z(z)\right) e^{ikx - i\omega t}$$

and $$\vec{B} = \left(0,\ B_y(z),\ 0\right) e^{ikx - i\omega t}$$

We can check the above ansatz by using Maxwell's eq^{n}s and they do satisfy!

Also, the system being isotropic and $\varepsilon(\omega)$ is not a tensor, hence the above assumption for $\vec{E}$ & $\vec{B}$ fields works! :)

From the last discussion class, we had the following component of fields calculated:–

$$E_x(z) = \underset{\substack{\uparrow \\ \text{Const}}}{E_0} \begin{cases} e^{-\sqrt{k^2 - \frac{\omega^2}{c^2}}\, z}, & z > 0 \\ e^{+\sqrt{k^2 - \frac{\omega^2}{c^2}\frac{\varepsilon(\omega)}{\varepsilon_0}}\, z}, & z < 0 \end{cases}$$

and $$B_y(z) = \begin{cases} \dfrac{-i\omega}{c^2\sqrt{k^2 - \omega^2/c^2}}\, e^{-z\sqrt{k^2 - \omega^2/c^2}}, & z > 0 \\ \dfrac{+i\omega\,\varepsilon(\omega)}{c^2\varepsilon_0\sqrt{k^2 - \frac{\omega^2}{c^2}\frac{\varepsilon(\omega)}{\varepsilon_0}}}\, e^{z\sqrt{k^2 - \frac{\varepsilon(\omega)}{\varepsilon_0}\left(\frac{\omega^2}{c^2}\right)}}, & z < 0 \end{cases}$$

TASK: To find $E_z(z)$, (can we find E_z using E_x?)

Ans: <u>Yes</u>.

Using the Boundary cond^n: $\nabla \cdot \vec{D} = 0$

$$\Rightarrow \nabla \cdot (\epsilon \vec{E}) = 0 \quad \text{OR} \quad \boxed{\nabla \cdot \vec{E} = 0}$$

<u>This gives us</u>

$$ik E_x + \frac{dE_z}{dz} = 0$$

$$\Rightarrow \boxed{E_z = -ik \int E_x \, dz}$$

E_z definitely has same functional dependence as of E_x.
We will be able to find the amplitude using this eq^n!

<u>For $z > 0$:</u> $E_x = E_0 e^{-\sqrt{k^2 - \frac{\omega^2}{c^2}} z}$

So $$E_z = \frac{-ik E_0 e^{-\sqrt{k^2 - \frac{\omega^2}{c^2}} z}}{-\sqrt{k^2 - \omega^2/c^2}}$$

$$\Rightarrow \boxed{E_z = \frac{ik E_0}{\sqrt{k^2 - \omega^2/c^2}} e^{-\sqrt{k^2 - \frac{\omega^2}{c^2}} z}} \quad \text{for } z > 0$$

<u>For $z < 0$:</u> $E_x = E_0 e^{\sqrt{k^2 - \frac{\omega^2}{c^2} \frac{\varepsilon(\omega)}{\varepsilon_0}} z}$

$$\Rightarrow \boxed{E_z = \frac{-ik E_0}{\sqrt{k^2 - \frac{\omega^2}{c^2} \frac{\varepsilon(\omega)}{\varepsilon_0}}} e^{\sqrt{k^2 - \frac{\omega^2}{c^2} \frac{\varepsilon(\omega)}{\varepsilon_0}} z}} \quad \text{for } z < 0$$

Looking at the boundary cond^{n}s we've used so far:-

1) $E_x(0+) = E_x(0-) \rightarrow$ Yes.

2) $B_y(0+) = B_y(0-) \rightarrow$ Yes.

3) $\epsilon_0 E_z(0+) = \varepsilon(\omega) E_z(0-) \rightarrow$ Yes, kind of redundant because its already satisfied in the above 2.

<u>To find $\omega = \omega(k)$ or $k = k(\omega)$</u> (Dispersion Relation)

Using B.C - 2 i.e $B_y(0+) = B_y(0-)$

<u>We get</u>

$$\frac{\varepsilon(\omega)}{\sqrt{k^2 - \frac{\varepsilon(\omega)}{\varepsilon_0}\frac{\omega^2}{c^2}}} = -\frac{\varepsilon_0}{\sqrt{k^2 - \frac{\omega^2}{c^2}}}$$

<u>Observations looking at above eq^{n}s</u>

1) We demand, $k > \frac{\omega}{c}$, because if not then we will get oscillatory behaviour in the Electric field along z-direc^n which we do not want.

2) Also, $\varepsilon(\omega) < 0$: One side of dielectrics should be have +ve permitivity & the other should have negative.

So, $$\frac{(\varepsilon(\omega))^2}{\left(k^2 - \frac{\varepsilon(\omega)}{\varepsilon_0}\frac{\omega^2}{c^2}\right)} = \frac{\varepsilon_0^2}{\left(k^2 - \omega^2/c^2\right)}$$

↓ Simplifying & expressing k, in terms of ω.

$$\Rightarrow \quad (\varepsilon(\omega))^2\left(k^2 - \frac{\omega^2}{c^2}\right) = \varepsilon_0^2\left(k^2 - \frac{\varepsilon(\omega)}{\varepsilon_0}\frac{\omega^2}{c^2}\right)$$

$$\rightarrow \quad \left(\frac{\varepsilon^2(\omega)}{\varepsilon_0^2} - 1\right)k^2 = \frac{\varepsilon(\omega)}{\varepsilon_0}\left(1 - \frac{\varepsilon(\omega)}{\varepsilon_0}\right)\omega^2/c^2$$

$$\Rightarrow \quad k^2 = \left(\frac{\frac{\varepsilon(\omega)}{\varepsilon_0} - \frac{\varepsilon^2(\omega)}{\varepsilon_0^2}}{\frac{\varepsilon^2(\omega)}{\varepsilon_0^2} - 1}\right)\omega^2/c^2$$

$$\Rightarrow \quad k = \pm\frac{\sqrt{\frac{\varepsilon(\omega)}{\varepsilon_0} - \frac{\varepsilon^2(\omega)}{\varepsilon_0^2}}}{\sqrt{\frac{\varepsilon^2(\omega)}{\varepsilon_0^2} - 1}}\,\frac{\omega}{c}$$

$$\Rightarrow \quad k = \frac{\omega}{c}\,\frac{\sqrt{\frac{\varepsilon(\omega)}{\varepsilon_0}}\cancel{\sqrt{\frac{\varepsilon(\omega)}{\varepsilon_0} - 1}}}{\sqrt{\frac{\varepsilon(\omega)}{\varepsilon_0} + 1}\cancel{\sqrt{\frac{\varepsilon(\omega)}{\varepsilon_0} - 1}}} = \frac{\omega}{c}\sqrt{\frac{\varepsilon(\omega)/\varepsilon_0}{\frac{\varepsilon(\omega)}{\varepsilon_0} + 1}}$$

$$\boxed{k = \frac{\omega}{c}\sqrt{\frac{\varepsilon(\omega)/\varepsilon_0}{\frac{\varepsilon(\omega)}{\varepsilon_0} + 1}}}$$

Upto this point, the derivation is general.

Now using the Drude Model

$$\frac{\varepsilon(\omega)}{\varepsilon_0} = 1 - \frac{\omega_p^2}{\omega^2} \quad ; \omega_p \rightarrow \text{Plasma frequency.}$$

So $k = \frac{\omega}{c}\sqrt{\frac{1-\omega_p^2/\omega^2}{2-\omega_p^2/\omega^2}}$

$$\boxed{k = \frac{\omega}{c}\sqrt{\frac{1-\omega_p^2/\omega^2}{2-\omega_p^2/\omega^2}}} \rightarrow (\star)$$

Dispersion Relation.

Remember: we're only looking at frequencies smaller than the plasma frequencies, otherwise 'k' would become imaginary & will give a decaying solution along x, where as $\vec{E}$ field needs to be oscillating along X. So $\boxed{\omega < \omega_p}$ (Energy is confined near the surface)

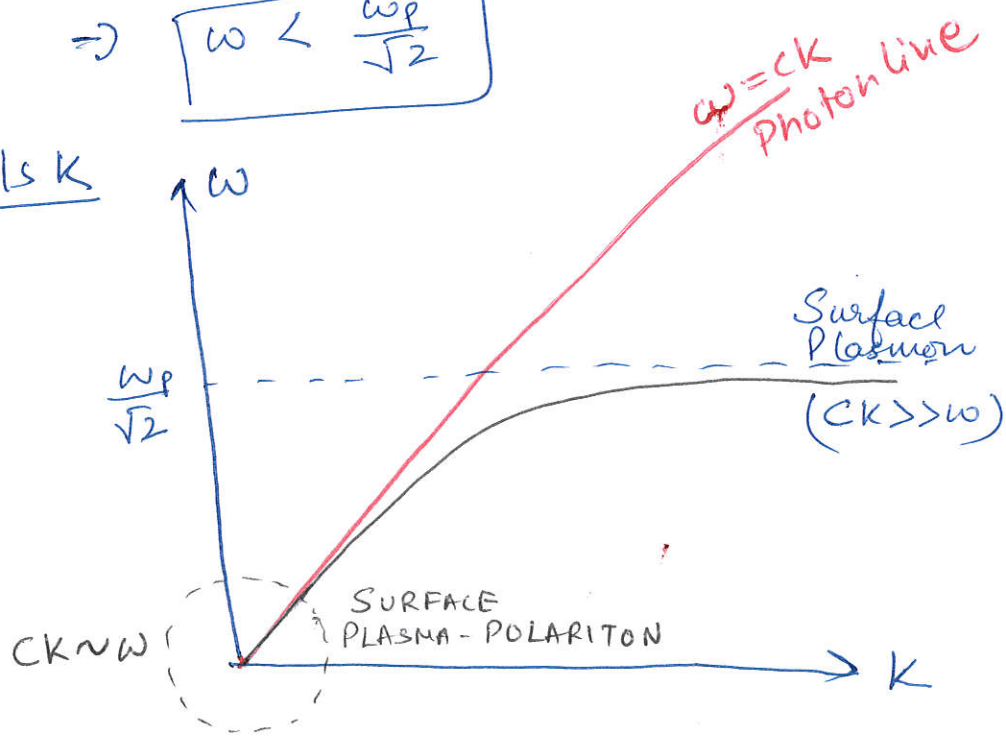
Also, looking at the denominator of $\star$ eqn,

$$2 - \frac{\omega_p^2}{\omega^2} > 0$$

$$\Rightarrow \boxed{\omega < \frac{\omega_p}{\sqrt{2}}}$$

Plot of ω vs k

- at large k, $\omega \longrightarrow \frac{\omega_p}{\sqrt{2}}$
- for small k $\omega \rightarrow 0$



Analysing the limits

(1) In limit $ck \gg \omega$, $E_x(z) \propto e^{ikx} e^{-kz}$ Wave is Confined in Plasmon region.

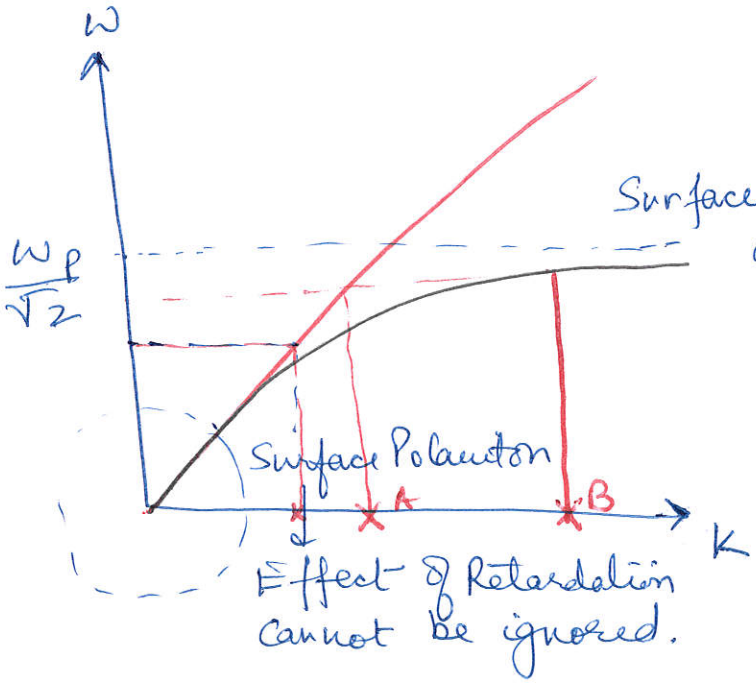
Shows, faster decay.

(2) In the limit $ck \sim \omega$, $e^{-\sqrt{k^2-\frac{\omega^2}{c^2}}\, z}$ almost goes to zero

$\Rightarrow$ Decay is very slow

i.e wave is least confined in the Polariton region

MORE COMMENTS



Surface Plasmon Region is almost as case of Electrostatics i.e No retardation.

Look at A & B

A → k_{light}

B → $k_{surface\ plasma\ polariton}$

$k_{light} < k_{spp}$

$\Rightarrow \lambda_{spp} < \lambda_{light}$

This indicates we get better resolution for SPP.

* Final Note.

We will derive $\omega = \frac{\omega_p}{\sqrt{2}}$ using electrostatics after the spring break.

Enjoy !!!